

Kiran Kunchala

Python Full Stack Engineer

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SUMMARY

Python Full Stack Engineer with 2+ years of experience designing and deploying end-to-end distributed applications using Python (FastAPI, Flask, Django), React.js, PostgreSQL, Docker, and AWS. Skilled in developing high-throughput asynchronous REST APIs, implementing modern single-page applications (SPAs) with Redux Toolkit and Material UI, and managing containerized CI/CD workflows. Strong background in database optimization, authentication architectures, and cross-functional Agile delivery.

SKILLS

Programming Languages Python (AsyncIO, OOP, Scripting), JavaScript (ES6+), Java, SQL, C++
Frontend Development React.js, Redux Toolkit, React Router, React Hooks, JavaScript (ES6+), HTML5, CSS3, Material UI, Axios
Backend Development FastAPI, Flask, Django, RESTful APIs, Pydantic, SQLAlchemy, Uvicorn, Microservices, Celery
Databases PostgreSQL, MySQL, SQLite, MongoDB, NoSQL, Relational Databases, Database Query Optimization, Indexing, Execution Plan Analysis, Query Tuning
Cloud and DevOps Docker, Kubernetes, Continuous Integration and Continuous Delivery (CI/CD), GitHub Actions, Git, AWS (EC2, S3)
Testing and Development Tools Pytest, Postman, Unittest, JIRA, OpenAPI/Swagger, Vite/Webpack
Security JSON Web Token (JWT) Authentication, Role-Based Access Control (RBAC), Structured Logging, Health Check APIs
Data & Machine Learning Pandas, NumPy, Scikit-learn, Streamlit, NLP Pipelines

EXPERIENCE

Software Developer Feb 2025 - Present
FLSmith

- Architected and developed 22+ end-to-end enterprise full-stack modules leveraging Python (FastAPI) on the backend and React.js on the frontend, reducing manual operational processing by 42%.
- Designed and integrated 25+ asynchronous RESTful APIs using FastAPI and Pydantic, bridging React single-page applications with PostgreSQL and slashing round-trip latency by 31%.
- Built 45+ accessible, reusable UI components using React.js, Material UI, and Redux Toolkit, establishing centralized state management and cutting frontend build time by 35%.
- Engineered secure JWT-based authentication and Role-Based Access Control (RBAC) across the full application stack, supporting 6 user roles and 4,800+ daily active enterprise users.
- Optimized full-stack performance through asynchronous backend I/O, database indexing, React code-splitting, and memoization, elevating UI benchmark scores from 74 to 92.
- Containerized multi-tier FastAPI and React services with Docker and coordinated container deployments via Kubernetes, achieving 99.9% uptime across production environments.
- Established automated health checks, error boundaries, and structured server logging to expedite full-stack bug tracking and defect resolution.
- Partnered with QA, UI/UX designers, and product owners in fast-paced Agile sprints to ship 32+ production features with a 95% sprint delivery rate.

Jr Software Engineer (Backend) Jul 2022 - May 2023
Wipro Ltd

- Engineered and maintained 18+ enterprise services using Python and Java, interfacing with internal UI dashboards to automate workflows and cut manual effort by 22%.
- Designed and documented 15+ secure RESTful endpoints, defining clean OpenAPI specifications to ensure seamless payload contract integration with client-side applications.
- Refactored complex relational queries and database schemas across MySQL and PostgreSQL, improving analytical reporting speeds by 33%.
- Collaborated closely with frontend developers and QA engineers to diagnose cross-stack edge cases, validate API contracts with Postman, and reduce production defects by 25%.
- Authored unit tests with Pytest and participated in routine code reviews, CI/CD runs, and Agile ceremonies to maintain clean, scalable software standards.

PROJECTS

Stock Data Processing System with Fault Tolerance Oct 2023 – Jan 2024

- Developed a cloud-native financial data platform utilizing Python (FastAPI), PostgreSQL, Docker, and AWS EC2 to process and visualize real-time market data.
- Engineered non-blocking asynchronous RESTful endpoints to ingest high-frequency data streams, increasing request throughput by 25%.
- Implemented resilience features including exponential backoff retries, fault recovery logic, and structured logging to maintain high service availability during simulated node downtime.
- Structured backend payloads for fast serialization, decreasing client-side data rendering latency by 20%.

NLP Sentiment & Emotion Classification Platform Jan 2023 – May 2023

- Created an end-to-end NLP web application using Python (Flask), Scikit-learn, and SQLite to analyze emotional sentiment from user text inputs.
- Built dynamic client-side input interfaces connected to Flask REST endpoints for real-time classification and visual confidence metrics.
- Streamlined data preprocessing pipelines and model inference logic, achieving a 20% reduction in inference latency and an 88% overall classification accuracy.

EDUCATION

Masters - Computer Science Aug 2023 - Dec 2024
Roosevelt University, Chicago

Bachelor of Technology - Computer Science Jun 2020 - May 2023
Tirumala Engineering College – JNTUK